

KEMEROVO, Russian Federation

WMO: 296420

Lat: 55.270N

Lon: 86.107E

Elev: 863

StdP: 14.24

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-23.9	-34.3	0.8	-29.2	-28.9	1.1	-23.7	26.8	15.1	23.4	14.2	6.0	160	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	84.2	66.4	80.8	64.9	78.3	63.7	69.3	79.0	67.4	76.7	65.7	74.9	8.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	99.3	73.6	64.2	92.8	72.0	62.4	87.1	70.4	33.9	79.0	32.3	76.9	31.0	74.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	20.7	18.6	DB	-33.0	89.4	7.9	2.5	-38.6	91.2	-43.2	92.7	-47.7	94.1	-53.4	95.9	(4)
(5)			WB	-32.7	72.9	8.0	2.9	-38.4	75.0	-43.1	76.7	-47.6	78.3	-53.3	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	1.5	7.0	21.5	37.0	51.7	62.6	66.3	61.4	49.7	36.4	18.4	6.8		
	DBStd	24.95	14.41	12.68	10.86	9.91	10.13	7.99	5.92	6.70	8.16	9.60	13.85	13.97		
	HDD50	6938	1504	1204	884	403	106	8	0	3	107	431	949	1339		
	HDD65	11103	1969	1624	1349	840	425	136	57	150	463	887	1399	1804		
	CDD50	1525	0	0	0	13	158	385	506	357	97	9	0	0		
	CDD65	214	0	0	0	0	12	63	98	39	2	0	0	0		
	CDH74	1700	0	0	0	1	169	531	672	294	33	0	0	0		
	CDH80	364	0	0	0	0	38	116	144	61	5	0	0	0		
	WSAvg	9.1	9.2	9.2	10.0	10.1	10.2	8.0	7.2	7.5	8.5	9.6	10.0	10.0		
	PrecAvg	21.9	1.2	1.0	1.0	1.1	1.7	2.7	3.0	2.7	1.9	1.9	1.9	1.8		
	PrecMax	27.5	2.2	2.2	2.6	2.2	3.6	5.1	4.9	5.7	3.7	3.3	2.9	3.1		
(15) Precipitation	PrecMin	15.8	0.1	0.0	0.2	0.1	0.4	1.0	0.4	1.1	0.4	1.1	0.8	0.4		
	PrecStd	2.9	0.6	0.5	0.6	0.5	0.8	1.1	1.1	1.0	0.9	0.6	0.6	0.7		
	0.4%	DB	33.8	34.1	46.4	69.5	84.6	87.6	88.0	86.1	78.9	65.8	46.8	35.4		
	2%	DB	28.7	30.4	41.3	62.8	78.6	83.3	84.2	80.8	71.7	59.2	42.6	31.8		
	5%	DB	25.1	28.2	37.9	57.4	73.7	80.3	80.7	77.0	67.3	54.3	37.8	28.4		
(23) Mean Coincident Wet Bulb Temperatures	10%	DB	21.3	24.6	35.6	51.9	69.5	76.8	77.4	73.5	62.6	49.9	34.4	24.9		
	MCWB	20.0	23.2	32.2	43.1	54.0	62.6	65.2	62.5	53.8	44.3	32.4	23.6	23.6		
	0.4%	WB	32.0	32.5	39.7	53.9	63.2	69.8	72.2	70.0	63.6	53.4	43.8	32.8		
	2%	WB	27.6	28.7	36.6	50.3	60.2	67.2	70.2	67.4	59.5	49.9	39.0	30.2		
	5%	WB	24.2	26.3	34.4	47.2	57.5	65.5	68.4	65.6	57.1	47.3	35.4	26.7		
(31) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	WB	20.2	23.1	32.2	43.6	55.0	63.7	66.9	63.7	54.7	44.4	32.5	23.7		
	MCDB	21.4	24.3	35.2	51.0	66.6	74.0	75.2	71.9	61.3	49.0	34.1	24.8	24.8		
	5% DB	MCDBR	12.9	14.0	14.9	23.6	25.9	22.9	21.0	22.0	23.6	20.9	12.0	13.5		
	5% WB	MCWBR	12.8	14.0	13.3	22.0	23.3	19.7	18.4	19.6	20.8	19.4	11.5	13.5		
	MCWBR	12.8	13.5	10.9	14.2	12.4	9.7	9.6	10.0	11.5	12.3	9.8	13.3	13.3		
(40) Clear-Sky Solar Irradiance	taub	0.292	0.326	0.375	0.402	0.406	0.403	0.429	0.406	0.371	0.355	0.326	0.296	0.296		
	taud	2.008	1.962	1.904	2.042	2.207	2.301	2.229	2.275	2.319	2.222	2.025	1.952	1.952		
	Ebn at Noon	186	224	241	257	265	267	257	255	248	217	167	148	148		
	Edh at Noon	22	34	46	46	42	38	41	36	30	26	22	19	19		
(44) All-Sky Solar Radiation	RadAvg	280	599	1027	1400	1602	1868	1690	1386	921	466	232	174	174		
	RadStd	31	43	96	88	186	162	119	108	113	53	40	29	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page